

Evaluating Structural Encodings and Systematic Zero-Embedding Ablations for GraphGPS

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Abstract

GraphGPS proposes a modular recipe for graph Transformers that combines positional and structural encodings (PE/SE), local message-passing, and global attention and demonstrates strong empirical performance across several benchmarks [1]. While the paper provides a taxonomy of many PE/SE choices and reports limited ablations including a “none” (no encoding) condition on a small set of datasets, it leaves a broader empirical evaluation of several candidate encodings to future work. This report formulates two precise empirical questions: (1) quantify the performance floor induced by removing all PE/SE via a systematic zero-embedding ablation across representative datasets, and (2) measure the marginal value and cost of selected “left-behind” encodings from the paper’s taxonomy (notably discrete Ricci curvature and cheap global graph statistics such as diameter, girth, and nodes/edges ratio) when added as plug-in SEs to the GraphGPS pipeline. We provide notation, concrete contrasts, and hypotheses grounded in expressivity theory and recent curvature analyses to motivate these experiments.

1 Introduction

Graph structured data are ubiquitous in domains such as molecular chemistry, social networks, and knowledge graphs. Two dominant modelling paradigms have emerged: local message-passing neural networks (MPNNs) that propagate information via edges, and Transformer-style global attention that allows long-range interactions. Rampášek et al. [1] synthesize these approaches into the GraphGPS recipe: (i) enrich inputs with positional/structural encodings (PE/SE), (ii) apply local MPNN blocks, and (iii) apply global attention blocks. The modular design and provided implementation enable systematic insertion and evaluation of new encodings.

GraphGPS’s taxonomy lists numerous candidate PE/SE types (global, relative, local) but explicitly states the authors empirically focus on a subset (e.g., Laplacian PE, RWSE) while leaving other encodings to future work [1]. The paper includes an ablation showing a “none” (no encoding) condition for a select few datasets; however, this zero-encoding entry is not reported uniformly across the wider benchmark suite. Thus, two practically important gaps remain: (i) a reproducible baseline quantifying how much the GraphGPS backbone alone contributes across a broader set of datasets, and (ii) an empirical evaluation of selected untested encodings from the taxonomy that are inexpensive to compute or theoretically motivated.

The planned investigation targets both gaps. We focus on two classes of left-behind encodings: (a) **local geometric SEs**, specifically discrete Ricci curvature (Ollivier and/or Forman variants), which quantify local neighborhood expansion/contracting behavior and have been linked to over-squashing and over-smoothing phenomena in message propagation [2]; and (b) **coarse global statistics** (diameter, girth, nodes/edges ratio, number of components) that cheaply summarize long-range geometry and cycle structure. These encodings are natural plug-ins to GraphGPS: curvature can be added as edge or aggregated node features, and global scalars can be broadcast as graph/node features or fed to the readout.

2 Problem Formulation

We now formalize the two experimental problems and the minimal contrasts required for reproducibility.

2.1 Notation

Let $G = (V, E)$ denote an attributed graph with node features $X \in \mathbb{R}^{|V| \times d}$. Let P denote a collection of positional/structural encodings (node/edge/graph level). In cases where an SE metric (e.g., Ricci curvature) is mathematically undefined for a specific substructure (such as an isolated edge), we initialize these missing attributes using a learnable zero-embedding, $\mathbf{0}_{emb}$, to maintain dimensional consistency without injecting false structural priors.

GraphGPS implements a parameterized mapping $f_\theta(X, P, E)$ trained to minimize supervised loss $\mathcal{L}(\theta) = \mathbb{E}_{(G, y)}[\ell(f_\theta(X, P, E), y)]$ on dataset $\mathcal{D}$. Partition the encoding sets as:

$$P_{\text{base}} = \{\text{encodings evaluated in GraphGPS (e.g., LapPE, RWSE, relative PE)}\}$$

$$P_{\text{new}} = \{\text{Ricci curvature (edge/node agg), diameter, girth, nodes/edges ratio}\}$$

2.2 Empirical questions

Q1 — Systematic zero-embedding ablation. Although GraphGPS reports a “none” condition in limited ablations, we ask: what is the true performance gap Δ_{zero} between the standard GraphGPS model with P_{base} and GraphGPS run with $P = \emptyset$ (utilizing the zero-embedding fallback) across a broader, representative benchmark set? Formally,

$$\Delta_{\text{zero}}(D) = \text{Perf}_D(f_\theta(P_{\text{base}})) - \text{Perf}_D(f_\theta(\emptyset))$$

measured per dataset D (with Perf being MAE, ROC-AUC, or Accuracy as appropriate). This quantifies the floor provided by the backbone alone and indicates where encodings are strictly essential.

Q2 — Marginal value and cost of left-behind SEs. For each $p \in P_{\text{new}}$ we define the marginal gain:

$$\Delta(p; D) = \text{Perf}_D(f_\theta(P_{\text{base}} \cup \{p\})) - \text{Perf}_D(f_\theta(P_{\text{base}}))$$

We also measure preprocessing time $T_{\text{prep}}(p; D)$, additional storage per graph, and per-epoch runtime overhead. Hypotheses:

- **H1:** Ricci curvature yields positive $\Delta(p; D)$ on datasets with pronounced bottlenecks/bridges or where message propagation is hampered.
- **H2:** Cheap global statistics (diameter, girth) yield modest $\Delta(p; D)$ and improve sample efficiency in low-data regimes, at negligible preprocessing cost for small graphs.

2.3 Minimal experimental contrasts

To keep the study focused and reproducible we propose these primary contrasts for each dataset D :

1. **Baseline:** $f_\theta(P_{\text{base}})$ (as in GraphGPS).
2. **Zero-Embedding:** $f_\theta(\emptyset)$.
3. **Single-SE:** $f_\theta(P_{\text{base}} \cup \{p\})$ for each $p \in P_{\text{new}}$.
4. **Combined:** $f_\theta(P_{\text{base}} \cup P_{\text{new}})$.

All models use the same training recipes and hyperparameters where possible; results report predictive metrics, preprocessing cost, and runtime overhead. For curvature, we distinguish Ollivier (more costly, transport-based) and Forman (faster, combinatorial) variants and report both cost and effect.

3 Literature Review

This review is compact and focused on grounding our empirical questions.

GraphGPS and encoding taxonomy. Rampášek et al. [1] formalize the GPS recipe and catalog many candidate PE/SE types. Their empirical evaluation concentrates on a subset (e.g., LapPE, RWSE, relative PE) and shows these provide marked improvements; they also include a “none” condition in limited ablations. The authors explicitly state that evaluating additional encodings is future work—a gap this project addresses by expanding the zero-embedding analysis and by testing selected left-behind SEs.

Expressivity and the role of encodings. Theoretical results tie the distinguishing power of standard MPNNs to the 1-WL test; without additional distinguishing features, MPNNs cannot separate certain non-isomorphic graphs [3, 4]. GraphGPS uses encodings to break such limitations empirically. Measuring Δ_{zero} thus indicates to what extent practical datasets depend on these augmentations versus the inductive power of the backbone.

Discrete curvature and message propagation. Discrete Ricci curvature (Ollivier, Forman) measures local geometry and has been connected to over-squashing and communication bottlenecks in GNNs [2]. Curvature-aware rewiring or curvature features have been proposed to mitigate bottlenecks, and prior work reports task-dependent benefits. GraphGPS lists curvature as a local SE candidate but does not evaluate it; our plug-in evaluation tests whether curvature adds value when combined with GraphGPS’s local/global modules.

Global graph statistics as cheap priors. Global scalars (diameter, girth, nodes/edges ratio) summarize coarse topology and are inexpensive on small graphs; they may inform whether stronger global attention is necessary or whether local motifs dominate. While simple, these descriptors are practical and understudied in the context of modern graph Transformer recipes.

Tooling and computational considerations. GraphRicciCurvature provides off-the-shelf implementations for Ollivier and Forman curvature on NetworkX graphs; Ollivier curvature can be computationally heavy (optimal transport steps) while Forman curvature is faster and often used in practice [5]. Reporting preprocessing cost alongside performance is therefore essential for practical recommendations.

References

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